INFO - 521
Introduction to Machine Learning
Final Project

German Bank Loan Defaulter Analysis

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1. Introduction:

The main objective of this project is to develop a machine learning model for the bank to predict whether the customer will default or not based on the historical data of various customers that did and did not default on their loan. It is crucial to predict loan defaulters as they can cause financial losses and affect the bank's ability to lend loans to other customers. Identifying loan defaulters helps banks perform risk management in an optimized manner. The German bank is facing issues with loan defaulters and wants to build a machine learning model to address the problem using a dataset that contains historical data of the customers who have taken loans from it. The dataset has the data of a thousand(1000) customers consisting of various attributes along with their respective default status indicating whether they defaulted on their loan or not.

The dataset has 17 columns and 1000 rows where the columns represent various attributes or features of the customers. Each row represents a customer along with their corresponding feature variables. Each customer is represented by the following 17 attributes:

a. 'checking_balance': It represents the amount of money available in the customer's checking account.
b. 'months_loan_duration': It represents the duration since the loan was taken in months.
c. 'credit_history': It represents the credit history of the customer.
d. 'purpose': It represents the reason for which the loan was taken.
e. 'amount': It represents the amount of loan taken.
f. 'savings_balance': It represents the savings account balance..
g. 'employment_duration': It represents the duration of employment.
h. 'percent_of_income': It represents the loan installment amount as the percent of customer's income.
i. 'years_at_residence': It represents the number of years the applicant has lived at their current residence.
j. 'age': It represents the customer's age.

k. ‘other_credit’: It represents the other existing credits.
l. ‘housing’: It represents the type of housing owned by the customer.
m. ‘existing_loans_count’: It represents the existing count of each customer’s loans.
n. ‘job’: It represents the customer’s job type.
o. ‘dependents’:It represents the number of dependents the customer has.
p. ‘phone’: It represents whether the customer has a phone or not.
q. ‘default’: It indicates whether the customer has defaulted on their loan or not.

The dataset has 17 columns out of which the ‘default’ column is the target or the response that needs to be predicted using the remaining variables or features. It is essential to check the number of categorical and numerical variables present in the dataset and in addition to this, each categorical and numerical variable’s distribution should be computed to get an overview of each feature’s distribution. It is also crucial to check the relationship between variables as it may help us in reducing the number of predictors required to train the machine learning models. Therefore, it is interesting to check the distribution of different variables along with the relationship between them.

2. Methods and Materials:

a. Exploratory Data Analysis(EDA):

The dataset has 1000 entries with 17 columns with no null values in any of the columns. Out of the 17 columns, 10 are categorical and the remaining 7 are numerical. Each of 10 columns consist of information regarding a categorical variable and each of the 7 columns consist of information about a numerical variable. The ten categorical variables are phone, housing, other_credit, job, savings_balance, purpose, checking_balance, credit_history, employment_duration and default. The seven numerical variables are months_loan_duration, amount, percent_of_income, years_at_residence, age, existing_loans_count and dependents. The categorical variable default is the response that needs to be predicted and the remaining variables are treated as predictors. The dataset has no missing values, null values and duplicate rows in it.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 17 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   checking_balance      1000 non-null   object
 1   months_loan_duration  1000 non-null   int64
 2   credit_history        1000 non-null   object
 3   purpose               1000 non-null   object
 4   amount                1000 non-null   int64
 5   savings_balance       1000 non-null   object
 6   employment_duration   1000 non-null   object
 7   percent_of_income     1000 non-null   int64
 8   years_at_residence    1000 non-null   int64
 9   age                   1000 non-null   int64
 10  other_credit          1000 non-null   object
 11  housing               1000 non-null   object
 12  existing_loans_count  1000 non-null   int64
 13  job                   1000 non-null   object
 14  dependents            1000 non-null   int64
 15  phone                 1000 non-null   object
 16  default               1000 non-null   object
dtypes: int64(7), object(10)
memory usage: 132.9+ KB
```

Fig-1: Dataset Information

Exploring Categorical Variables:

In this section, each categorical variable is explored by plotting pie-charts and count-plots. There are 700 non-defaulters and 300 defaulters in the dataset indicating the presence of class imbalance in the dataset. The dataset is biased towards customers that did not default. Out of all the customers, 596 possess a phone whereas 404 do not. Based on the job type there are 630 skilled workers, 200 unskilled workers, 148 management workers and 22 unemployed customers. 713 customers own a house whereas 179 customers live in a rented home. Surprisingly, 108 customers neither own a house nor live in a rented home. Around 814 customers did not take credit from anywhere but 139 customers took other credit from a bank and 47 customers took credit from a store.

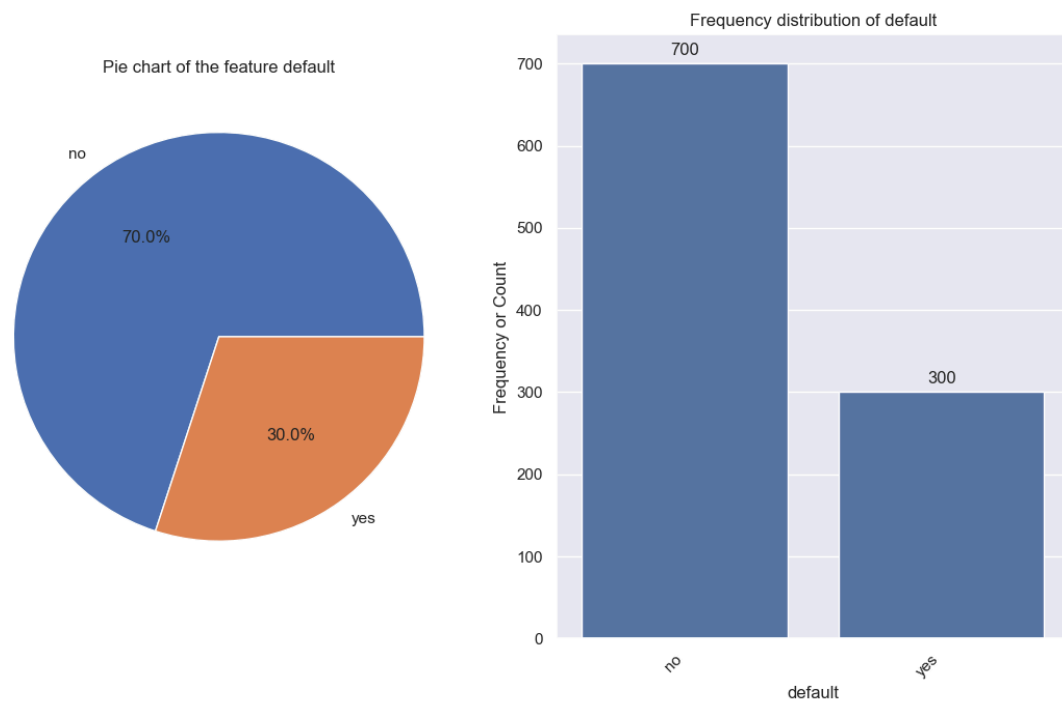

Fig-2: Countplot of the target or response default

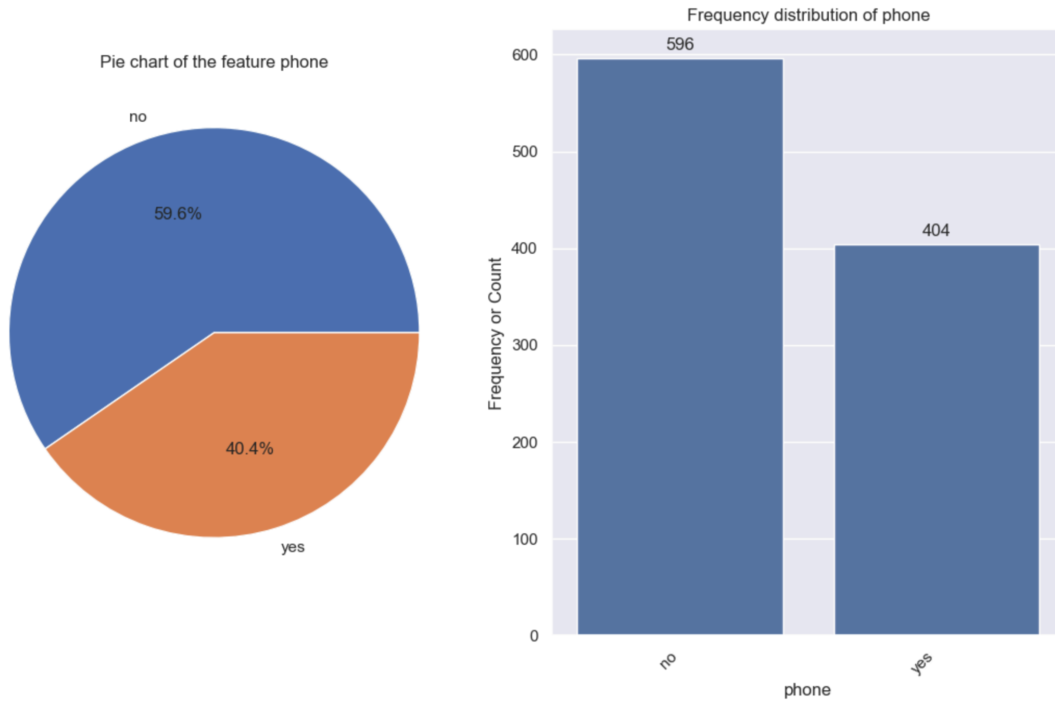

Fig-3: Countplot of predictor phone

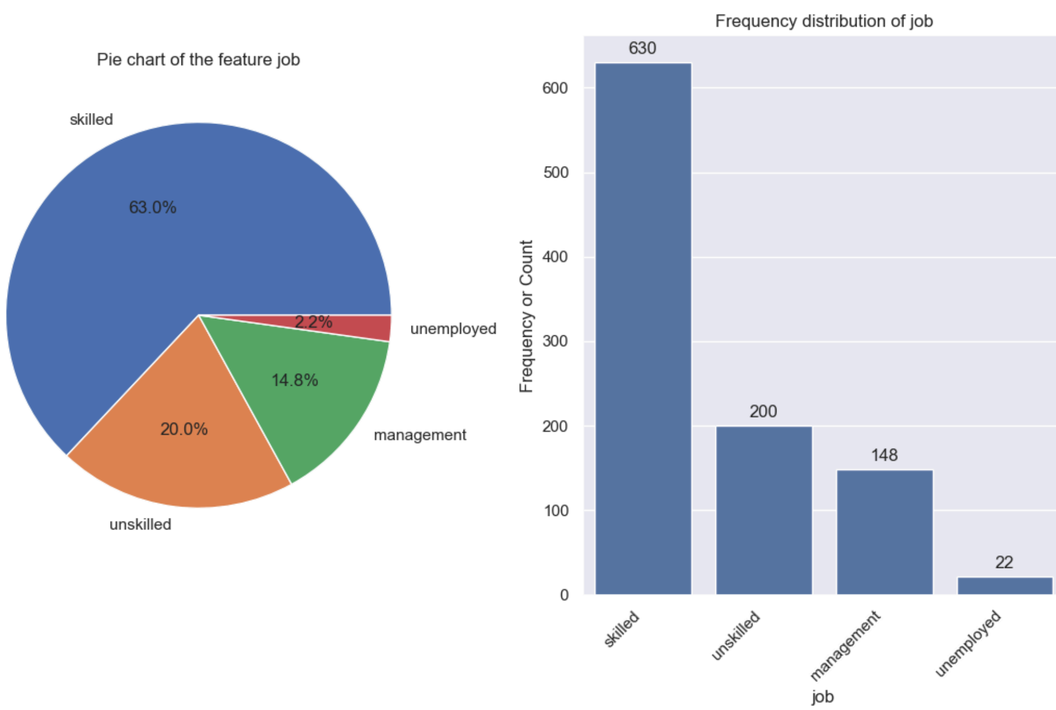

Fig-4: Countplot of predictor job

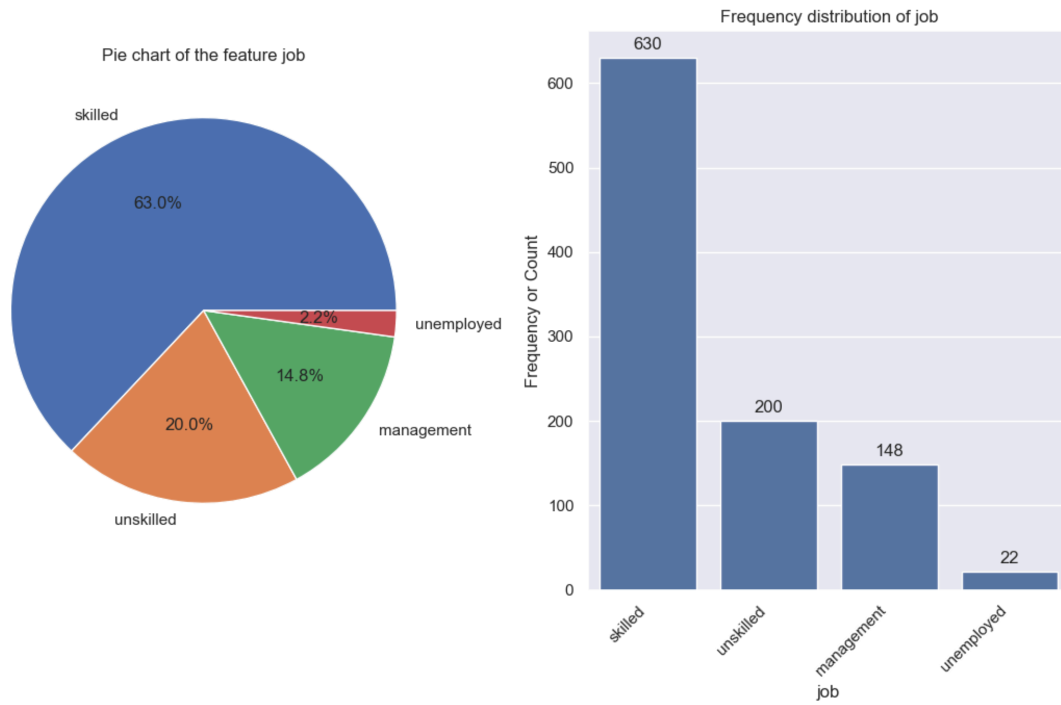

Fig-5: Countplot of predictor job

3. Results:

4. Discussion:

5. Conclusions: